

Homework 1: DDL — March 31, 2026, Midnight

Python Programming (You can use AI for help)

1 Lab # 1 (3 Pts)

2 Lab # 2 (3 Pts)

Questions for Tech Report (6 Pts, **≤ 3 Pages**, You can use AI for help)

- Given N labeled samples $\{x_n \in \mathbb{R}, y_n \in \mathbb{R}\}_{i=1}^N$, we build a D -order polynomial regression model with data-dependent noise assumption, i.e.,

$$y = \sum_{d=1}^D x^{d-1} \theta_d + \epsilon, \text{ with } p(\epsilon|x) = \frac{1}{\sqrt{2\pi x^2}} \exp\left(-\frac{\epsilon^2}{2x^2}\right), \quad (32)$$

- 1 Learn the model θ via MLE, and derive its close-form solution. (4 Pts)
- 2 In addition, suppose that the elements of θ obey Laplace distributions, i.e.,

$$\theta_d \sim \text{Exp}(0, b_d) \Leftrightarrow p(\theta_d) = \frac{1}{b_d} \exp\left(-\frac{\theta_d}{b_d}\right), \text{ and } \theta_d \geq 0, \forall d = 1, \dots, D. \quad (33)$$

Learn the model θ via MAP, and derive its learning process. (4 Pts)

Delayed submission: -1 Pts per week.